

COSC2148/COSC3130 Computing Research and Project Preparation

Assignment 2

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Section 1

Introduction

1.1 DELETE original question

Introduction - 1 page (2 marks)

Briefly describe your topic and why it is important or interesting. You could start by describing the area of interest and later summarise the existing solutions and their limitations. You could also explain the specific focus of your research and why you are doing so. Feel free to include specific information, such as detailed statistics, that will back up your claims (and thus make your statements much stronger).

In summary, an introduction should:

- Introduce the topic and its significance.
- Give background and context.
- Outline your problem and research questions.

Some important questions to guide your introduction include:

- Who is interested in the topic (e.g. scientists, practitioners, policymakers, particular members of society)?
- How much is already known about the problem?
- What is missing from current body knowledge?
- What new insights will your research contribute?
- Why is this research worth doing

1.2 aaaSection2

1.2.1 aaaSection2Subsection1

aaa

Section 2

Research Questions

2.1 DELETE original question

Research questions - 1 page (4 marks)

This section should address the technical issues you will be addressing in your research. It is about the “what” (you will be addressing/focussing when you start your thesis).

Describe here the specific research questions to be addressed. You need to break the problem you are addressing into specific sub-problems (called here research questions) that could help you during the thesis to come up with innovative solutions properly.

Show how the research questions are interesting and worth addressing.

Please include at least two (2) research questions, but you are limited to a maximum of three (3) research questions.

For each research question, describe what it is, its significance, and why it is important to address it.

2.2 aaaSection2

2.2.1 aaaSection2Subsection1

aaa

Section 3

Literature Review

3.1 DELETE original question

Literature Review - 2 pages (7 marks)

It is indeed essential to demonstrate that you are familiar with the most important research on your selected topic. A strong literature review convinces the reader that your research/project has a solid foundation in existing knowledge or theory. It also shows that you are not simply repeating what others have already done or said.

The aim of this section is to show exactly how your project will contribute to discussions in the field.

- Compare and contrast: what are the main theories, methods, debates, and controversies?
- Be critical: what are the strengths and weaknesses of different approaches?
- Show how your research fits in: how will you build on, challenge, or synthesise the work of others?

3.2 aaaSection2

3.2.1 aaaSection2Subsection1

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Section 4

Research Methodologies

4.1 DELETE original question

methodologies - 1 page (3 marks)

This section should address the “how”, that is, how you will be addressing your research (or research questions).

The aim of this section is to convince the reader that your overall research design and analysis methods will enable you to solve the research problem (i.e., research questions) you have identified and also enable you to accurately and effectively interpret the results of your research. Consequently, it is important that the research design and methods section is well-written, clear, and logically organised.

Specifically, this section should be clearly tied to the specific objectives of your study; however, it is also important to draw on and include examples from the literature review that relate to your design and intended methods. In other words, you must clearly demonstrate how your study utilises and builds on previous studies as it relates to the research design and intended methods. For example, what methods have other researchers used in similar studies?

Finally, this section should clearly show the different pieces that you will need to do in your project (in the form of a table). Break it down and give a timeline. Allow at least four (4) weeks to write the thesis.

4.2 aaaSection2

4.2.1 aaaSection2Subsection1

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Section 5

Evaluation

5.1 DELETE original question

evaluation - 1 page (3 marks)

This section is important and should clearly show the evaluation plan of your research. This evaluation plan could be used later during your thesis to clearly demonstrate that you made research contributions in the field of study.

This section will need to address the following points:

- Benchmarking: clearly select some existing methods/algorithms you will need to compare your research work with. It is important to clearly explain these methods/algorithms as well as justify the reasons for choosing them.
- Data sets: describe at least two well-accepted data sets that you will be using to compare your research work against the selected benchmarked methods/algorithms. Again, you will need to justify the reasons for selecting such data sets.
- Metrics: to compare your research work with benchmarked methods/algorithms, you clearly need to list the metrics to be used for this purpose. This could be, for example, "execution time" if you are comparing the performance of your work with other methods/algorithms. You also need to explain why you have selected such metrics. List at least two metrics.

5.2 aaaSection2

5.2.1 aaaSection2Subsection1

aaa

Section 6

Project Management

6.1 DELETE original question

project management - 1 page (1 mark)

This section should outline how you plan to manage your research project throughout the semester.

It is essentially a Project Plan that clearly defines the scope and deliverables, including explicit exclusions.

It should present a structured schedule with milestones and a transparent critical path, while assigning responsibility to specific individuals to ensure accountability.

It must also incorporate risk management to anticipate threats and formulate appropriate responses, protecting project objectives from avoidable failure.

This section should also clearly state the methods you will use to manage the project, such as Agile sprints, Scrum ceremonies, Kanban, etc., and specify the enabling tools (e.g., MS Project, Trello, JIRA, Slack, etc.) that will operationalise these methods.

Briefly explain how these methods and the supporting tools together will facilitate scheduling, progress tracking, communication, and reporting. Specifically, you must include the following:

- High-level scope and deliverables of your research project. Clearly state exactly what you will deliver and explicitly rule out what you will not.
- Timeline with major milestones derived from the Work Breakdown Structure (WBS). This should reflect the schedule of your research project and include any critical paths. Note: you don't need to include the WBS itself.
- Clear assignment (ownership) of each task to project team member(s), not roles, for accountability and monitoring.
- Approach in risk management, identify risks and proposed mitigation and contingency plan, as well as escalation paths. List the major risks, rank them, and state what you will do before, when, and after they occur.

6.2 aaaSection2

6.2.1 aaaSection2Subsection1

aaa

Bibliography

Appendix A

Generative AI Usage

Declaration

[REPLACE THIS LINE WITH YOUR GENERATED DECLARATION FROM
<https://aiattribution.github.io/>]

Tool Report: [Tool Name, e.g. ChatGPT / RMIT Val / Copilot]

Field	Details
1. GenAI tool used	[e.g. ChatGPT 4o, RMIT Val, GitHub Copilot]
2. Purpose for this assignment	[e.g. Brainstorming research questions, drafting outline, rewriting for clarity, code generation]
3. Prompts or instructions given	[Include key prompts used. If conversation was lengthy, include all your prompts in full and use [...] to truncate AI responses only.]
4. Number of prompts given	[e.g. 7]
5. Summary of GenAI output	[Describe what the AI produced. Include brief excerpts where relevant.]
6. Verification of GenAI output	[Explain how you checked the AI’s output for factual accuracy, source validity, and logical soundness. e.g. “All cited sources were manually located and verified via Google Scholar.”]
7. Usage or modification of output	[Explain how you used, adapted, or discarded the AI-generated content in your final submission.]
8. Reflection	[What worked well? What did not? How did it support your learning? What value did it add?]

Table A.1: Generative AI usage report — [Tool Name]